

- 8** In a roller coaster ride, a train of mass 1200 kg and carrying a load of 400 kg is pulled up a smooth slope at a constant speed of 1.5 m s^{-1} by an electric motor to a height of 80.0 m as shown in Fig. 8.1. The train with its load is then released from rest at point A and moved to point B along the curved track.

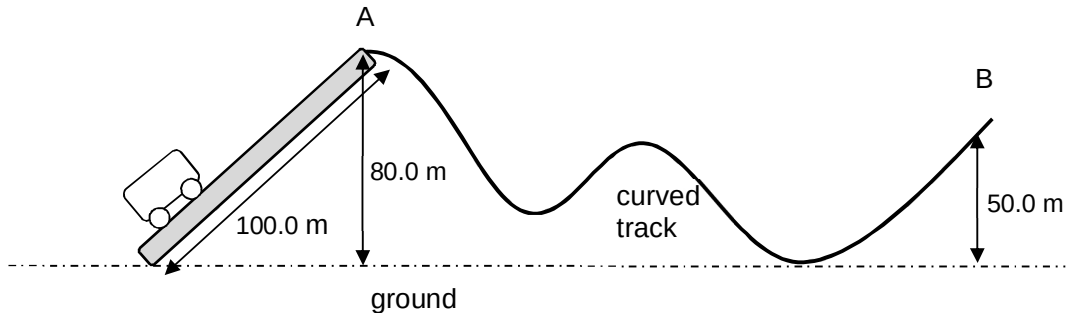


Fig. 8.1

(a) (i) Calculate the minimum force required to pull the train with its load up the slope at the given speed.

minimum force =	N	[2]
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Assuming no friction,
The upslope pull needs to balance the downslope component of weight.

Minimum Force required = $m g \sin \theta$
 $= (1200 + 400) \times 9.81 \times 80.0 / 100.0$
 $= 12556.8 \text{ N} = 12600 \text{ N (3 s.f.)}$

(ii) The efficiency of the motor is 80%. Calculate the electrical power that is supplied to the motor to pull the train with its load up the slope to point A.

		electrical power = kW	[3]
		Output Power = $F v$ $= 12556.8 \times 1.5$ $= 18835 \text{ W}$ Actual power required by motor $= \text{Output Power} / \text{efficiency}$ $= 18835 / 0.80$ $= 23544 \text{ W} = 24 \text{ kW} \text{ (2 s.f.)}$	
	(iii)	Between A and B, the train experiences an average resistive force of 3.0 kN. If the train arrives at B with a speed of 8.0 m s^{-1} , determine the distance travelled by the train between A and B.	
		distance travelled = m	[3]
		By conservation of energy, from A to B, Loss in GPE = Gain in KE + Work done against resistive force $mg(h_A - h_B) = (\frac{1}{2} mv_B^2 - 0) + 3000 \times d$ $(1200 + 400)(9.81)(80.0 - 50.0) = \frac{1}{2} (1200 + 400)(8.0^2) + 3000d$ $d = 139.89 = 140 \text{ m} \text{ (2 s.f.)}$	
	(iv)	Suggest why the resistive force on the train is unlikely to remain constant. (Ignore the resistive force due to air resistance.)	
			
			
			
			[1]
		A component of resistive force is the frictional force between the wheels and the platform. <u>Normal contact force exerted by the track on the wheels differs in magnitude at different points of the ride, which causes the frictional force exerted by the track on the wheels to vary as well.</u>	
	(b)	In another segment of the ride, the train loop the loop in a circular track as shown in Fig.8.2. The radius of the circular motion is 25 m. Point D is at the bottom of the circular motion. Point E is at the right of the circular motion. Point F is at the top of the circular motion. Point G is at the left of the circular motion.	

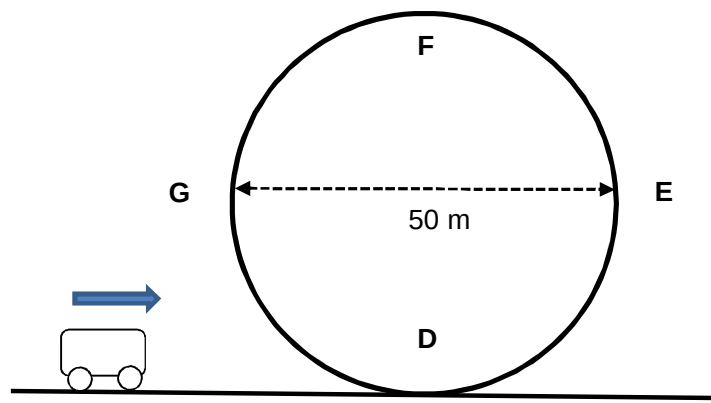


Fig. 8.2

- (i) On Fig. 8.3 below, draw the free body diagram of the train at point D and point F.

Point D



Point F



Fig. 8.3

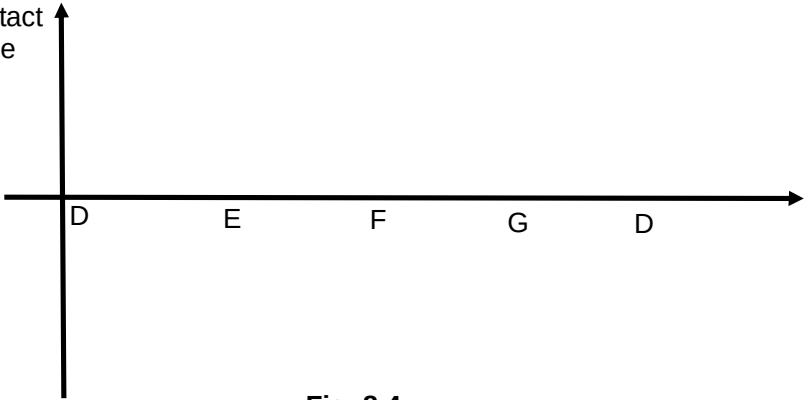
[2]

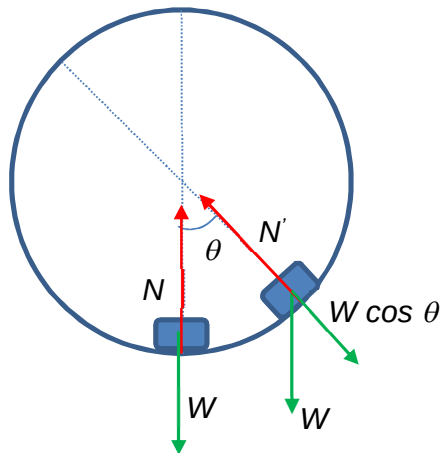
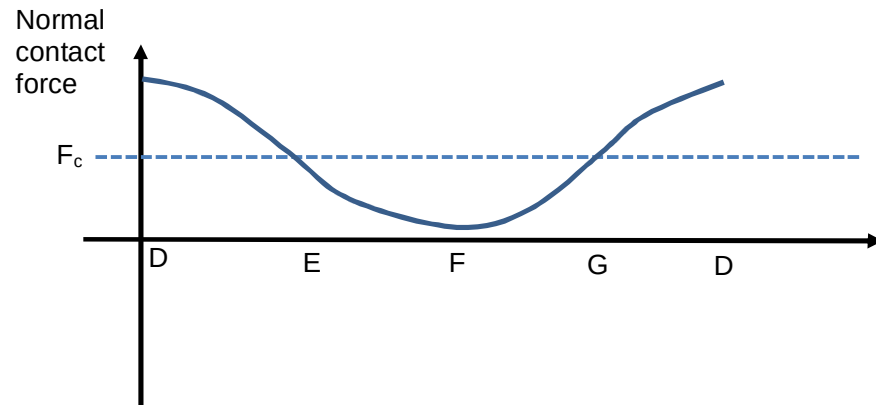
Weight directed downwards for both top and bottom.
 Normal contact Force (at bottom, D) directed upwards.
 Normal contact Force (at Top, F) directed pointing downwards.

1 mark – Labelled and named all forces in the correct directions.
 1 mark – Normal contact force at D **greater than** Weight, and Weight constant in magnitude at D and at F.

- (ii) Determine the minimum speed of the train at point D so that it can loop the loop.

At the top of the circular motion,
 Normal contact force and train's weight provide for the centripetal force required:
 $N + mg = F_c$
 $N = F_c - mg$
 Normal contact force at top needs to be > 0 for the train to stay on track.
 $F_c - mg > 0$

			$\frac{mv_F^2}{r} - mg > 0$ $\frac{v_F^2}{r} - g > 0$ $v_F^2 > rg$ $v_F > \sqrt{25 \times 9.81}$ $v_F > 15.660 \text{ m s}^{-1}$ Total KE at bottom = min KE at top + GPE gained $\frac{mv_D^2}{2} = \frac{mv_F^2}{2} + mgh$ $\frac{v_D^2}{2} = \frac{v_F^2}{2} + gh$ $v_D^2 = v_F^2 + 2gh$ $v_D^2 = (15.660)^2 + 2(9.81)(50)$ $v_D = 35.018 = 35 \text{ m s}^{-1}$	
			Minimum speed = m s ⁻¹	[4]
	(c)	It is suggested to keep the speed of the train constant.		
		(i)	On Fig. 8.4, sketch the graph to show how the normal contact force experienced by the train varies as it moves from Point D to G and back to D. Normal contact force  Fig. 8.4	
				[3]

Solution

General Formula:

$$N = F_c + mg \cos \theta$$

Point D : $\theta = 0^\circ$

$$N = F_c + mg$$

Point E : $\theta = 90^\circ$

$$N = F_c$$

Point F : $\theta = 180^\circ$

$$N = F_c - mg$$

Point G : $\theta = 270^\circ$

$$N = F_c$$

Where F_c is constant throughout since m , v , r remain constant.

All positive values. [Deduct 1 mark if there are negative values]

Point D and F correct [B1] : accept $N \geq 0$ at F.

Point E and G correct (shows indication that N is equal at these 2 points)

[B1]

Sinusoidal shape. [B1]

(ii) Discuss why it is difficult to maintain the train at constant speed during the loop

				
				
				[2]
		L3	On the upward part of the journey, the train <u>requires an engine to do positive work / supply energy to it at varying rates</u> because kinetic energy (KE) remains unchanged while gravitational potential energy (GPE) needs to increase at <u>varying rates</u>. On the downwards part of the journey, the train <u>requires a braking system to do negative work / dissipate energy from it at varying rates</u> to prevent increase in KE, since GPE converts to KE <u>at varying rates</u>.	

9	(a)	Define <i>electric potential</i> .
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